

3.8 Factoring Special Polynomials

LESSON FOCUS

Investigate some special factoring patterns.



Make Connections

The area of a square plot of land is one hectare (1 ha).

$$1 \text{ ha} = 10\,000 \text{ m}^2$$

So, one side of the plot has length $\sqrt{10\,000} \text{ m} = 100 \text{ m}$

Suppose the side length of the plot of land is increased by x metres.

What binomial represents the side length of the plot in metres?

What trinomial represents the area of the plot in square metres?

Construct Understanding

THINK ABOUT IT

Work with a partner.

You may need algebra tiles.

■ Determine each product.

$$(x + 1)^2$$

$$(x + 2)^2$$

$$(x + 3)^2$$

$$(x - 1)^2$$

$$(x - 2)^2$$

$$(x - 3)^2$$

$$(2x + 1)^2$$

$$(3x + 1)^2$$

$$(4x + 1)^2$$

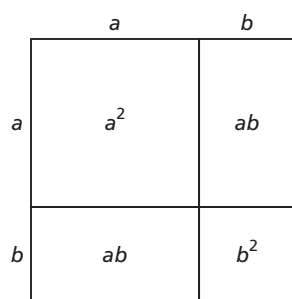
$$(2x - 1)^2$$

$$(3x - 1)^2$$

$$(4x - 1)^2$$

- What patterns do you see in the trinomials and their factors above?
How could you use the patterns to factor these trinomials?
 $4x^2 + 20x + 25$ $9x^2 - 12x + 4$
- Write two more polynomials that have the same pattern,
then factor the polynomials.
Write a strategy for factoring polynomials of this type.

Here is a square with side length $a + b$:



$$\begin{aligned}
 \text{Its area is: } (a + b)^2 &= (a + b)(a + b) \\
 &= a(a + b) + b(a + b) \\
 &= a^2 + ab + ab + b^2 \\
 &= a^2 + 2ab + b^2
 \end{aligned}$$

We say that $a^2 + 2ab + b^2$ is a **perfect square trinomial**.

$$\begin{array}{lcl}
 \text{The square of the 1st term in the binomial} & & \text{Twice the product of the first and second terms in the binomial} \\
 \downarrow & \downarrow & \downarrow \\
 \text{The square of the 2nd term in the binomial} & & \\
 (a + b)^2 = a^2 + 2ab + b^2 & &
 \end{array}$$

Perfect Square Trinomial

The area model of a perfect square trinomial results in a square.

In factored form, a perfect square trinomial is:

$$a^2 + 2ab + b^2 = (a + b)(a + b), \text{ or } (a + b)^2$$

$$\text{And, } a^2 - 2ab + b^2 = (a - b)(a - b), \text{ or } (a - b)^2$$

We can use these patterns to factor perfect square trinomials.

Example 1**Factoring a Perfect Square Trinomial**

Factor each trinomial. Verify by multiplying the factors.

a) $4x^2 + 12x + 9$ b) $4 - 20x + 25x^2$

SOLUTION

a) $4x^2 + 12x + 9$

Arrange algebra tiles to form a rectangle.

The rectangle is a square with side length $2x + 3$.

$$4x^2 + 12x + 9 = (2x + 3)^2$$

To verify, multiply:

$$\begin{aligned}(2x + 3)(2x + 3) &= 2x(2x + 3) + 3(2x + 3) \\ &= 4x^2 + 6x + 6x + 9 \\ &= 4x^2 + 12x + 9\end{aligned}$$

Since this trinomial is the same as the original trinomial, the factors are correct.

b) $4 - 20x + 25x^2$

The 1st term is a perfect square since $4 = (2)(2)$

The 3rd term is a perfect square since $25x^2 = (5x)(5x)$

The 2nd term is twice the product of $5x$ and 2 :

$$20x = 2(5x)(2)$$

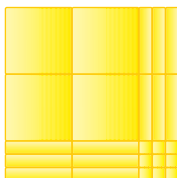
Since the 2nd term is negative, the operations in the binomial factors must be subtraction.

So, the trinomial is a perfect square and its factors are:

$$(2 - 5x)(2 - 5x), \text{ or } (2 - 5x)^2$$

$$\begin{aligned}\text{To verify, multiply: } (2 - 5x)(2 - 5x) &= 2(2 - 5x) - 5x(2 - 5x) \\ &= 4 - 10x - 10x + 25x^2 \\ &= 4 - 20x + 25x^2\end{aligned}$$

Since this trinomial is the same as the original trinomial, the factors are correct.

**CHECK YOUR UNDERSTANDING**

1. Factor each trinomial. Verify by multiplying the factors.

a) $36x^2 + 12x + 1$

b) $16 - 56x + 49x^2$

[Answers: a) $(6x + 1)^2$
b) $(4 - 7x)^2$]

When $ax^2 + bx + c$ is a perfect square trinomial, how are a , b , and c related?

In Lesson 3.7, you multiplied two binomials in two variables. We can use the inverse process to factor trinomials of this form. We can use the strategy of logical reasoning or decomposition.

Example 2**Factoring Trinomials in Two Variables**

Factor each trinomial. Verify by multiplying the factors.

a) $2a^2 - 7ab + 3b^2$ b) $10c^2 - cd - 2d^2$

SOLUTION

a) $2a^2 - 7ab + 3b^2$

Use logical reasoning. Since the 3rd term in the trinomial is positive, the operations in the binomial factors will be the same. Since the 2nd term is negative, these operations will be subtraction.

The two binomials will have the form: $(?a - ?b)(?a - ?b)$

List the possible factors of $2a^2$ and $3b^2$, then use guess and test to determine which combination of products has the sum of $-7ab$.

$$\begin{array}{r} 1a \quad -1b \\ \diagdown \quad \diagup \\ 2a \quad -3b \end{array}$$

$$\begin{array}{r} 1a \quad -3b \\ \diagdown \quad \diagup \\ 2a \quad -1b \end{array}$$

Form the products,
then add.

$$-3ab - 2ab = -5ab \quad -1ab - 6ab = -7ab \quad \leftarrow \text{This is the correct combination of factors.}$$

So, $2a^2 - 7ab + 3b^2 = (2a - b)(a - 3b)$

To verify, multiply:

$$\begin{aligned} (2a - b)(a - 3b) &= 2a(a - 3b) - b(a - 3b) \\ &= 2a^2 - 6ab - ab + 3b^2 \\ &= 2a^2 - 7ab + 3b^2 \end{aligned}$$

Since this trinomial is the same as the original trinomial, the factors are correct.

b) $10c^2 - cd - 2d^2$

Use decomposition. Since the 3rd term in the trinomial is negative, the operations in the binomial factors will be addition and subtraction.

The two binomials will have the form: $(?c + ?d)(?c - ?d)$

The product of the coefficients of c^2 and d^2 is: $10(-2) = -20$

Write $-1cd$ as a sum of two terms whose coefficients have a product of -20 .

(Solution continues.)

CHECK YOUR UNDERSTANDING

2. Factor each trinomial. Verify by multiplying the factors.

a) $5c^2 - 13cd + 6d^2$

b) $3p^2 - 5pq - 2q^2$

[Answers: a) $(5c - 3d)(c - 2d)$

b) $(3p + q)(p - 2q)$]

Factors of -20	Sum of Factors
1, -20	$1 - 20 = -19$
$-1, 20$	$-1 + 20 = 19$
2, -10	$2 - 10 = -8$
$-2, 10$	$-2 + 10 = 8$
4, -5	$4 - 5 = -1$
$-4, 5$	$-4 + 5 = 1$

The two coefficients are 4 and -5 , so write the trinomial $10c^2 - cd - 2d^2$ as $10c^2 + 4cd - 5cd - 2d^2$.

Remove a common factor from the 1st pair of terms, and from the 2nd pair of terms:

$$10c^2 + 4cd - 5cd - 2d^2 = 2c(5c + 2d) - d(5c + 2d)$$

Remove the common binomial factor.

$$\text{So, } 10c^2 - cd - 2d^2 = (5c + 2d)(2c - d)$$

To verify, multiply:

$$\begin{aligned} (5c + 2d)(2c - d) &= 5c(2c - d) + 2d(2c - d) \\ &= 10c^2 - 5cd + 4cd - 2d^2 \\ &= 10c^2 - cd - 2d^2 \end{aligned}$$

Since this trinomial is the same as the original trinomial, the factors are correct.

Another example of a special polynomial is a **difference of squares**.

A difference of squares is a binomial of the form $a^2 - b^2$.

We can think of it as a trinomial with a middle term of 0.

That is, write $a^2 - b^2$ as $a^2 + 0ab - b^2$.

Consider the binomial $x^2 - 25$.

This is a difference of squares because $x^2 = (x)(x)$ and $25 = (5)(5)$.

We write $x^2 - 25$ as $x^2 - 0x - 25$.

To factor this trinomial, look for 2 integers whose product is -25 and whose sum is 0.

The two integers are 5 and -5 .

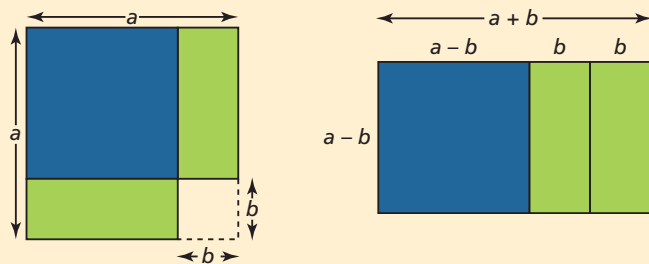
$$\text{So, } x^2 - 25 = (x + 5)(x - 5)$$

This pattern is true for the difference of any two squares.

Difference of Squares

A difference of squares has the form $a^2 - b^2$.

In factored form, $a^2 - b^2 = (a - b)(a + b)$



Example 3 Factoring a Difference of Squares

Factor each binomial.

a) $25 - 36x^2$ b) $5x^4 - 80y^4$

SOLUTION

a) $25 - 36x^2$

Write each term as a perfect square.

$$\begin{aligned} 25 - 36x^2 &= (5)^2 - (6x)^2 && \text{Write these terms in} \\ & && \text{binomial factors.} \\ &= (5 + 6x)(5 - 6x) \end{aligned}$$

b) $5x^4 - 80y^4$

As written, each term of the binomial is not a perfect square. But the terms have a common factor 5. Remove this common factor.

$$\begin{aligned} 5x^4 - 80y^4 & \\ = 5(x^4 - 16y^4) & \quad \text{Now write each term in the} \\ & \quad \text{binomial as a perfect square.} \end{aligned}$$

$$= 5[(x^2)^2 - (4y^2)^2] \quad \text{Write these terms in binomial factors.}$$

$$= 5(x^2 - 4y^2)(x^2 + 4y^2) \quad \text{The 1st binomial is also a difference of squares.}$$

$$= 5(x + 2y)(x - 2y)(x^2 + 4y^2)$$

CHECK YOUR UNDERSTANDING

3. Factor each binomial.

a) $81m^2 - 49$

b) $162v^4 - 2w^4$

[Answers: a) $(9m - 7)(9m + 7)$
b) $2(3v - w)(3v + w)(9v^2 + w^2)$]

Does a sum of squares factor?
Explain.

Discuss the Ideas

1. How do the area models and rectangle diagrams support the naming of a perfect square trinomial and a difference of squares binomial?
2. Why is it useful to identify the factoring patterns for perfect square trinomials and difference of squares binomials?
3. Why can you use the factors of a trinomial in one variable to factor a corresponding trinomial in two variables?

Exercises

A

4. Expand and simplify.

a) $(x + 2)^2$	b) $(3 - y)^2$
c) $(5 + d)^2$	d) $(7 - f)^2$
e) $(x + 2)(x - 2)$	f) $(3 - y)(3 + y)$
g) $(5 + d)(5 - d)$	h) $(7 - f)(7 + f)$
5. Identify each polynomial as a perfect square trinomial, a difference of squares, or neither.

a) $25 - t^2$
b) $16m^2 + 49n^2$
c) $4x^2 - 24xy + 9y^2$
d) $9m^2 - 24mn + 16n^2$
6. Factor each binomial.

a) $x^2 - 49$	b) $b^2 - 121$
c) $1 - q^2$	d) $36 - c^2$

B

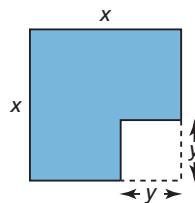
7. a) Factor each trinomial.

i) $a^2 + 10a + 25$	ii) $b^2 - 12b + 36$
iii) $c^2 + 14c + 49$	iv) $d^2 - 16d + 64$
v) $e^2 + 18e + 81$	vi) $f^2 - 20f + 100$

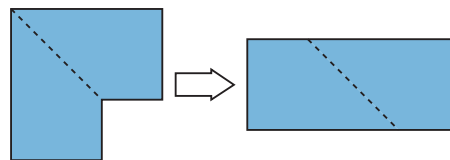
 b) What patterns do you see in the trinomials and their factors in part a? Write the next 4 trinomials in the pattern and their factors.
8. Factor each trinomial. Verify by multiplying the factors.

a) $4x^2 - 12x + 9$	b) $9 + 30n + 25n^2$
c) $81 - 36v + 4v^2$	d) $25 + 40h + 16h^2$
e) $9g^2 + 48g + 64$	f) $49r^2 - 28r + 4$

9. a) Cut out a square from a piece of paper. Let x represent the side length of the square. Write an expression for the area of the square. Cut a smaller square from one corner. Let y represent the side length of the cut-out square. Write an expression for the area of the cut-out square. Write an expression for the area of the piece that remains.



- b) Cut the L-shaped piece into 2 congruent pieces, then arrange as shown below.



- What are the dimensions of this rectangle, in terms of x and y ?
 What is the area of this rectangle?
- c) Explain how the results of parts a and b illustrate the difference of squares.

10. Factor each binomial. Verify by multiplying the factors.

a) $9d^2 - 16f^2$ b) $25s^2 - 64t^2$
 c) $144a^2 - 9b^2$ d) $121m^2 - n^2$
 e) $81k^2 - 49m^2$ f) $100y^2 - 81z^2$
 g) $v^2 - 36t^2$ h) $4j^2 - 225h^2$

11. Factor each trinomial.

a) $y^2 + 7yz + 10z^2$ b) $4w^2 - 8wx - 21x^2$
 c) $12s^2 - 7su + u^2$ d) $3t^2 - 7tv + 4v^2$
 e) $10r^2 + 9rs - 9s^2$ f) $8p^2 + 18pq - 35q^2$

12. Factor each trinomial. Which trinomials are perfect squares?

a) $4x^2 + 28xy + 49y^2$ b) $15m^2 + 7mn - 4n^2$
 c) $16r^2 + 8rt + t^2$ d) $9a^2 - 42ab + 49b^2$
 e) $12h^2 + 25hk + 12k^2$ f) $15f^2 - 31fg + 10g^2$

13. Factor.

a) $8m^2 - 72n^2$ b) $8z^2 + 8yz + 2y^2$
 c) $12x^2 - 27y^2$ d) $8p^2 + 40pq + 50q^2$
 e) $-24u^2 - 6uv + 9v^2$ f) $-18b^2 + 128c^2$

14. A circular fountain has a radius of r centimetres. It is surrounded by a circular flower bed with radius R centimetres.

- a) Sketch and label a diagram.
 b) How can you use the difference of squares to determine an expression for the area of the flower bed?
 c) Use your expression from part b to calculate the area of the flower bed when $r = 150$ cm and $R = 350$ cm.

15. a) Find an integer to replace $\square$ so that each trinomial is a perfect square.

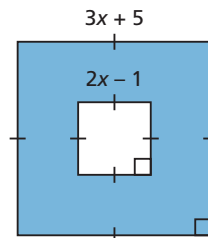
i) $x^2 + \square x + 49$
 ii) $4a^2 + 20ab + \square b^2$
 iii) $\square c^2 - 24cd + 16d^2$

- b) How many integers are possible for each trinomial in part a? Explain why no more integers are possible.

16. Find consecutive integers a , b , and c so that the trinomial $ax^2 + bx + c$ can be factored. How many possibilities can you find?

17. Use mental math to determine $(199)(201)$. Explain your strategy.

18. Determine the area of the shaded region. Simplify your answer.



C

19. a) Identify each expression as a perfect square trinomial, a difference of squares, or neither. Justify your answers.

i) $(x^2 + 5)^2$
 ii) $-100 + r^2$
 iii) $81a^2b^2 - 1$
 iv) $16s^4 + 8s^2 + 1$

- b) Which expressions in part a can be factored? Factor each expression you identify.

20. Factor fully.

a) $x^4 - 13x^2 + 36$
 b) $a^4 - 17a^2 + 16$
 c) $y^4 - 5y^2 + 4$

21. Factor, if possible. For each binomial that cannot be factored, explain why.

a) $8d^2 - 32e^2$
 b) $25m^2 - \frac{1}{4}n^2$
 c) $18x^2y^2 - 50y^4$
 d) $25s^2 + 49t^2$
 e) $10a^2 - 7b^2$
 f) $\frac{x^2}{16} - \frac{y^2}{49}$

Reflect

Explain how a difference of squares binomial is a special case of a trinomial. How is factoring a difference of squares like factoring a trinomial? How is it different? Include examples in your explanation.